S-11

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester 2021
Course: Econ1211 (Engineering Economics)
(Session: 2020-2021)
Time: 2 Hours **Full Marks: 40**
[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section A

1. (a) What do you mean by engineering economics? Why is engineering economics important for engineers? 5
   (b) What is the difference between simple interest and compound interest. 5

2. (a) What are the relationships between capital recovery factor and sinking fund factor? 3
   (b) Create cash flow diagrams from the perspective of the lender and the borrower. 2
   (c) Three different bank loan rates for electric generation equipment are listed below. Determine the effective rate on the basis of the compounding period for each rate. 5
      (a) 9% per year, compounded quarterly.
      (b) 9% per year, compounded monthly.
      (c) 4.5% per year, compounded weekly.

3. (a) Define the concepts of nominal interest rate and effective interest rate. 3
   (b) What amount invested in a savings account that pay 8% interest annually is worth $215,892.50 after 10 years? 4
   (c) What is effective interest rate for continuous compounding? 3

Section B

4(a) What do you mean by inflation? 3
   (b) Distinguish between demand-pull and cost-push inflation. 3
   (c) How much money must be invested in a retirement plan each month to accumulate $500,000 in 5 years? Assume an annual interest rate of 6% compounded monthly. 4

5(a) Discuss benefit cost ratio with a practical example. 5
   (b) A person is planning a new business. The capital outlay and cash flow pattern for the new business are listed below. The expected life of the business is five years. Find the rate of return for the new business. 5

Period	0	1	2	3	4	5
Cash flow	-1,00,000	30,000	30,000	30,000	30,000	30,000

6(a) What is value added tax? 2
   (b) Consider the following cash flow diagram: n = 30 years, F at 30 years = $+50,000, i = 4% per year compounded annually, A = $+2500 per year, inflation rate f = 2.5% per year. Calculate the present worth (PW) (i) without taking inflation into consideration, and (ii) taking inflation into consideration. 4
   (c) You deposit $1000 each year in an account for 5 years. At the time of the last deposit the bank tells you the account is worth $6,500. What rate of return did you earn on your investment? 4

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Even Semester 2021
Course: Econ1211 (Engineering Economics)
(Session: 2019-2020)
Time: 2 Hours **Full Marks: 35**
[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section A

1. (a) What do you mean by engineering economics? Discuss the importance of studying engineering economics as an engineer. 4.75
   (b) Distinguish between simple interest and compound interest. 4

2. (a) What is a geometric gradient series? Draw cash flow diagram of increasing and decreasing geometric gradient series. 4.75
   (b) What do you mean by sinking fund? A company desires to deposit 100,000 Taka in order to change a machine after 8 years. If annual rate of interest is 7% and it is counted twice a year, how much the company should deposit at the end of every six months? 4

3. (a) Define the concepts of nominal interest rate and effective interest rate. 2
   (b) What amount invested in a savings account that pay 8% interest annually is worth $215,892.50 after 10 years? 4
   (c) What is effective interest rate for continuous compounding? 2.75

Section B

4(a) Give practical applications of various interest formulas. 2
   (b) If Laurel made a $30,000 investment in a friend's business and received $50,000, 5 years later, determine the rate of return. 4
   (c) A person wishes to have a future sum of Tk. 1,00,000 for his son's education after 10 years from now. What is the single-payment that he should deposit now so that he gets the desired amount after 10 years? The bank gives 15% interest rate compounded annually. 2.75

5(a) Discuss the advantages and disadvantages of the benefit cost (B/C) ratio. 3
   (b) What is the B/C ratio for a firm considering an investment in a new manufacturing technology: 5.75
   Investment = $2,500,000 Net annual saving: $600,000
   Salvage value: $150,000 Project life = 10 years MARR (i%) = 15%

6(a) What are gross income, taxable income, average tax rate, and net profit after tax? 5
   (b) An individual submits a tax return to the revenue board. During the year his job provided him an income of $82,000. He had the first child during the year, and he plans to use the standard deduction of $9500 applicable for the year. Dividends and interest amounted to $3550, and an investment in a stock mutual fund had capital gains of $2500. Compute his average income tax if the tax rate is as follows: 3.75

Income ($)	Tax rate
0 – 14000	10%
14001 – 56800	15%
56801 – 114650	25%
More than 114650	30%

University of Rajshahi
Dept. of Computer Science and Engineering
B.Sc. Engg. Part-I, Even Semester, Examination 2020
Course Code: ECON 1211 Course Title: Engineering Economics

Time: 2 Hours **Full Marks: 35**

[N.B. Answer FOUR questions taking at least TWO from each Section.]

Section A

1. (a) Define engineering economy. Why is it important for engineers to study engineering economy? 4
   (b) Distinguish, with an example, between simple and compound interest rates. 4.75

2. (a) What is cash flow? Draw cash flow diagrams from both a borrower's point of view and a lender's point of view. 2
   (b) Explain the concept of arithmetic gradient with figures. 2
   (c) A construction firm pays $5,000.00 for the first year of a maintenance service contract that increases by $1,000.00 in each subsequent year until year six. If the interest rate is 6%, calculate the present worth of the uniform annual series and the gradient. 4.75

3. (a) Distinguish between nominal interest rate and effective interest rate. 2
   (b) Three different bank loan rates for electric generation equipment are listed below. Determine the effective rate on the basis of the compounding period for each rate. 4.75
      i) 9% per year, compounded quarterly.
      ii) 9% per year, compounded monthly.
      iii) 4.5% per 6 months, compounded weekly.
   (c) What is effective annual interest rates? 2

Section B

4. (a) What is the primary purpose of a public sector project? 1
   (b) Explain how public projects are different from private projects. 3
   (c) Distinguish between the conventional B/C ratio and the modified B/C ratio for a single project. 4.75

5. (a) What do you mean by inflation? How does inflation happen? 4
   (b) Derive and show the relationship between market interest rate, real interest rate and the inflation rate. 2
   (c) Money in a medium-risk investment makes a guaranteed 8% per year. Inflation rate has averaged 5.5% per year. What is the real rate of return on the investment? 2.75

6. (a) What are the main elements of after tax cash flows? 2
   (b) A medium-sized profitable corporation may buy a $15,000 used pickup truck for use by the shipping and receiving department. During the truck's 5-year useful life, it is estimated the firm will save $4000 per year after all the costs of owning and operating the truck have been paid. Truck salvage value is estimated at $4500. 6.75
      i) What is the before-tax rate of return?
      ii) What is the after-tax rate of return on this capital expenditure? Assume straight-line depreciation.

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. (Engg.) Part-I Even Semester Examination 2019
Course: ECON 1211 (Economics)
Full Marks: 35 **Time:** 2 Hours

[Answer any Four questions taking two from each section]

Section A

1. a) Define economics. Distinguish between microeconomics and macroeconomics. 4
   b) What are the central economic problems of an economy? How these problems are solved in different economic systems? 4.75

2. a) Define demand and supply. 2
   b) Derive demand and supply curves from corresponding demand and supply schedules. 4
   c) What is elasticity of demand? Distinguish between arc elasticity and point elasticity with examples. 2.75

3. a) What are the characteristics of perfect competition and monopoly markets? 3
   b) Draw TC, AC, MC and TFC curves. 3
   c) Define market equilibrium and present market equilibrium graphically. How is equilibrium restored in a market? 2.75

Section B

4. a) Distinguish between GDP and GNP with example. 3
   b) What is meant by Aggregate Demand (AD)? 1.75
   c) What is double counting problem? Give an example. 4

5. a) Explain the concepts of frictional, structural and cyclical unemployment with example. 3
   b) What is Philips curve? Draw short-run and long-run Philips curves. 3
   c) What is tax? Distinguish between direct tax and indirect tax with example. 2.75

6. a) What is a development plan? Write the principal characteristics of a development plan? 3
   b) Write the phases of development plan in Bangladesh since independence. 4
   c) What are the advantages of planning approach? 1.75

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering Part-1 EVEN Semester, Examination - 2018

Course Code: **ECON-1211** Course Title: **Economics**

Full Marks: 35 Time: 2 Hours

[N.B.: Answer **Four** (04) Questions taking any **Two** (02) Questions from **each Section**]



Section-A

1.(a) Define economics system. What are the different economic system? 2.00

(b) What are the features of capitalist economic system? Discuss its advantages and disadvantages. 6.75

2. (a) Explain demand and supply in economics. What are laws of demand and supply? 2.75

(b) What is equilibrium in the market? Explain with necessary figures. 4.00

(c) Consider a commodity $Q_{dt} = 18 - 3p$ and $Q_{st} = -3 + 4p$, where Q_{dt}, Q_{st} and p denotes quantity demanded, quantity supply and price respectively. Find the equilibrium price and quantity for that commodity. 2.00

3. (a) Define production function. 2.25

(b) Discuss all the factors of production in brief. 3.00

(c) Write the characteristics of a perfectly competitive market. 3.50

Section-B

4. (a) What is unemployment? Explain the different types of unemployment in brief. 5.25

(b) Explain the nature of unemployment in under-developed countries. 3.50

5. (a) What is inflation? Explain the demand-pull and cost- push inflation with example. 3.50

(b) Discuss in brief the causes of inflation. 5.25

6. (a) What is government budget? 2.00

(b) Distinguish between revenue and capital budget. 3.00

(c) What are the major sources of revenue and heads of expenditure of Bangladesh government? 3.75

University of Rajshahi

Department of Computer Science of Engineering

B. Sc. (Engg.) Part-I, Even Semester Exam - 2017

Course Title: ECON-1211 (Economics)

Total marks: 35 Time: 2 Hours

[Answer two questions from each section]

Section-A

1.	a)	Define Economics with criticism. Discuss the subject matter of Economics.	3
	b)	What are the central economic problems of every society? Discuss.	2.25
	c)	Explain the fundamental characteristics of different economics systems.	3.5
2.	a)	Distinguish between demand schedule and demand curve. Draw a demand curve from a hypothetical demand schedule.	3
	b)	What is meant by shift in demand and supply curves? Mention the factors that cause shifts in demand and supply curves.	2.75
	c)	Define elasticity of demand. Write down the formulae for computing price, income and cross elasticities of demand.	3
3.	a)	What is meant by production? Discuss the factors of production.	3
	b)	What is meant by cost of production? Distinguish between fixed costs and variable costs with the help of necessary figures.	3
	c)	What is a market in Economics? Discuss the characteristics of perfectly competitive market.	2.75

Section-B

4.	a)	What does economic growth mean?	2
	b)	Explain the difference between economic growth and economic development.	4
	c)	Explain the circular flow of income in an economy.	2.75
5.	a)	Describe the different methods of national income accounting.	5.75
	b)	Discuss the problems of national income accounting in Bangladesh.	3
6.	a)	What is meant by budget? How is a deficit budget financed?	3
	b)	Critically examine the development philosophy of the planning commission of Bangladesh.	3
	c)	Explain the major objectives of the 7th Five Year Plan of Bangladesh.	2.75

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. Engg. Part1 Even Semester, Examination-2016
Course: ECON-1211 (Economics)
Full Marks: 35 **Time: 2 Hours**
Answer two questions from each part

Professor Dr. Shaikh Shahidul Islam
Dept. of Computer Science & Engineering
University of Rajshahi.

Part A

1. (a) Define economics. What are the fundamental problems of any economy? 4
(b) Discuss the subject maters of microeconomics and macroeconomics. 2
(c) Explain the relation between economics and engineering 2.75

2. a) State the law of demand. Draw a demand curve using the demand equation $Q_d = 50-5p$ 3
3.75
b) What is supply? What are the causes of change in supply? Explain. 2
c) Using the following demand and supply equations, find the equilibrium quantity and market price: $Q_d = 20-2p$, $Q_s = 10+3p$

3. a) What is production? Discuss the four factors of production. 3
b) Define cost. Distinguish between fixed cost and variable cost. 3
c) What is a market? Discuss the main differences between perfect competition and monopoly. 2.75

Part B

4. (a) Define inflation and unemployment. 2
(b) How are inflation and unemployment related in the short-run? Explain this relationship with the help of Philips curve. 3
(c) What is business cycle? Discuss the phases of business cycle. 3.75

5. (a) What is national income? Distinguish between Gross Domestic Product (GDP) and Gross National Product (GNP). 3

(b) What do you mean by multiplier? Prove that the investment multiplier is $\frac{\Delta Y}{\Delta I} = \frac{1}{1-MPC}$ 3.75
(c) Suppose, the level of autonomous investment (I) in any economy is Tk. 200 crores and consumption function of the economy is $C = 80+0.75Y$. 2
(i) What will be the equilibrium level of income?
(j) What will be the increase in national income if the investment increases by Tk. 25 crores?

6. (a) What do you mean by budget? Distinguish between revenue budget and capital budget. 3
(b) What are the major sources of revenue of the government of Bangladesh? Discuss. 3
(c) Explain the major objectives of the first five years plan of Bangladesh. 2.75

University of Rajshahi

Department Of Computer Science and Engineering

B.Sc.(Engg.) Part-I, Even Semester Examination-2014

Course: ECON 1211 (Economics)

Full Mark: 35 Duration: 2 hours

[N.B. Answer any four questions taking two from each group. Figures in the right margin indicate full marks]

Part-A

1. a) What do you mean by economics? Explain the concepts of total product, average product and marginal product in economics. 3
   b) Mention the three stages of production. Explain why a rational producer concentrates producing on the second stage of the total product curve. 5.75

2. a) Discuss what are meant by price elasticity, cross price elasticity and income elasticity of demand. 3
   b) If the demand for the commodity y increases from 5 units to 8 units as a result of increase in income of a consumer from Tk. 50,000.00 to Tk. 65,000.00, find the income elasticity of demand and comment on the nature of the commodity. 2
   c) Show that the price elasticity of demand ranges from zero to infinity. 3.75

3. a) What is meant by a perfectly competitive market? State its assumptions. 3
   b) Explain the conditions for obtaining equilibrium of a firm under a perfect competition. 5.75

Part-B

4. a) Distinguish between GDP and GNP. 3
   b) Graphically explain the circular flow of national income in a simple two sector economy. 3.75
   c) Discuss various steps of business cycle. 2

5. a) What is inflation? 2
   b) What are the features and causes of inflation? 3.75
   c) Compare among the methods of measuring national income. 3

6. a) What is meant by multiplier? 2
   b) Describe the simple Keynesian model for the determination of national income with the help of expenditure method. 4
   c) What are the differences between revenue budget and capital budget? 2.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-I (Even Semester) Exam - 2013
Course: ECON 1211 (Economics)
Full Marks: 35 **Time:** 2 Hours
(Answer 4 questions taking any 2 from each group. Figures in the margin indicate full marks)

Group A

1.(a) Why is "what to produce" a problem in every economy? 2

(b) How does the price mechanism solve this problem in a free-enterprise economy? 3

(c) Define economic resources. What are the factors of production? Are they economic resources? 3.75

2.(a) State the law of demand and the law of supply. 3

(b) The following table refers the change in an individual's consumption of lemons and tea at home when the price of lemons rises (*ceteris paribus*) 2.75

	Before		After	
	Price (cents/unit)	Quantity (units/month)	Price (cents/unit)	Quantity (units/month)
Lemons	10	20	20	15
Tea	20	40	20	35

i) Draw a figure showing these changes
ii) Explain the figure drawn.

(c) Define ATC, AVC and MC. Draw them in a single diagram. 3

3.(a) What is meant by market equilibrium? What will happen in a market if the current price is above the equilibrium price? 3

(b) What happens to the equilibrium price in a market if the supply curve shifts to the left? 2.75

(c) Given the market demand function $QD_x = 120{,}000 - 20{,}000P_x$ and market supply function $QS_x = 20{,}000P_x$ 3
i) Find the equilibrium price and equilibrium quantity.
ii) Is the equilibrium condition stable? Why?
iii) Define unstable equilibrium and metastable equilibrium.

Group B

4.(a) Mention some macroeconomic variables. Is Rajshahi City Corporation's annual expenditure a macroeconomic variable? 3.75

(b) What does the CPI measure? How is the CPI different from the GDP deflator? 5

5.(a) Describe the major problems of GDP measurement. 4

(b) What problems does the underground economy pose for GDP measurement? 4.75

6.(a) Define aggregate output. 3

(b) Show the circular flow of national income for the three-sector model and derive the equation of national income from there. 5.75

Department of Computer Science and Engineering

University of Rajshahi

B.Sc. (Engg) Part-1 Examination 2012 Semester: Even

Course Title: Economics ourse Code: ECON 1211

Full Marks: 35 Time 3 Hours

(Answer any Four question taking two from each group)

Part A

1(a)	Critically explain the definition of Economics given by L. Robbins.	3
(b)	Discuss the central economic problems of a society.	2.75
(c)	What do you mean by economic law? What are the limitations of economic laws?	3
2(a)	State the law of demand. Distinguish between demand schedule and demand curve.	2.75
(b)	Distinguish between change in demand and change in quantity demand.	3
(c)	What is 'cross elasticity' of demand? How do you identify the relationship between two commodities with the help of 'cross elasticity' of demand?	3
3(a)	Distinguish between total fixed cost and total variable cost and present both in a single diagram	275
(b)	What is meant by equilibrium of a firm? State the conditions of the equilibrium of firm.	3
(c)	Discuss short run equilibrium of a firm under perfect competition.	3

Part B

4(a)	What is input-out analysis? State the assumptions of input-output analysis.	3
(b)	Define co-efficient matrix and technology matrix.	2
(c)	Suppose, there are n number of different industries in the economy. If the co-efficient matrix and final demand vector are given, then how can you obtain correct level of output for the industries?	3.75
5(a)	Define macroeconomics. State the main macroeconomic variables.	2.75
(b)	Discuss the circular flow of economic activities in a three sector economy.	3
(c)	Define unemployment and discuss different types of unemployment.	3
6(a)	Discuss how equilibrium national income is determined in a simple Keynesian two sector economy.	3.75
(b)	Define budge. Distinguish between revenue and development budget.	2.5
(c)	State the objectives and core targets of the Sixth Five-Year plan of Bangladesh.	2.5

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year 2nd Semester 2011
Course: ECON 1211 (Economics)
Time: 3 Hrs. Full Marks: 35
[N.B. Answer FOUR questions taking at least TWO from each part.]

PART-A

1.	(a) Define Economics.	2.75
	(b) Distinguish between Micro and Macro economics.	3
	(c) Discuss the central economic problems of every society.	3
2.	(a) What is demand? Explain the law of demand.	3
	(b) Why does a demand curve slope downward to the right?	2.75
	(c) What is elasticity of demand? From the demand function, Q=50-2P, find price elasticity of demand when P=8.	3
3.	(a) What is monopoly? What are the characteristics of a monopoly market?	2
	(b) What is meant by equilibrium of a firm? Derive the conditions of equilibrium of a firm.	3.75
	(c) Explain the short run equilibrium of a firm under perfect competition.	3

PART-B

4.	(a) What do you mean by input-output analysis? Mention the assumptions of input-output analysis.	3
	(b) Distinguish between open and closed input-output model.	2
	(c) Explain how the correct level of output is obtained in an input-output model?	3.75
5.	(a) Graphically present the circular flow of macroeconomic activity.	3
	(b) Distinguish between Gross Domestic Product and Gross National Product.	2.75
	(c) What are the problems of measuring national income?	3
6.	(a) What is budget? Distinguish between revenue budget and capital budget.	3
	(b) Distinguish between direct tax and indirect tax.	2.75
	(c) What are the major sources of revenue and major heads of expenditure of the government of Bangladesh?	3